

Micro Set 8. Government Intervention — Model Answer Scheme

Question 1

[2 marks]

$$\% \Delta = \frac{148 - 247}{247} \times 100\% = -40.1\%$$

[1 mark working, 1 mark answer.] Cigarette sales fell by approximately 40 per cent between 2018 and 2024.

Question 2

[2 marks]

An indirect tax is a tax levied on the producer or seller of a good, with the seller collecting the tax on behalf of the government from buyers as part of the transaction. [1 mark.]

Reason from Extract 1.1: tobacco generates “negative externalities through second-hand smoke and the public-health cost of smoking-related disease,” so the excise serves to internalise these external costs. [1 mark.]

Question 3

[4 marks]

Why tobacco tax raises revenue but barely reduces smoking prevalence.

Extract 1.1 reports that “tobacco demand is sufficiently inelastic that even substantial tax increases reduce quantity demanded modestly.” Inelastic demand has a specific revenue implication: when $PED < 1$ (in absolute value), a price rise generates a percentage fall in quantity smaller than the percentage rise in price, so total expenditure (and government revenue) rises. The 2022 excise increase generated S\$1.4 billion in additional annual revenue, confirming this. [2 marks for *PED + revenue mechanism.*]

However, the gap between sales (down 40%) and smoking rate (down only 1.8 percentage points, from 11.6% to 9.8%) reveals that existing smokers reduced *intensity* (cigarettes per smoker) but did not quit entirely. This pattern reflects nicotine addiction: quitting requires more than financial disincentive — it requires medical support, social networks, and time. [2 marks for *behavioural explanation.*]

Examiner Note. The distinction between intensive margin (cigs per smoker) and extensive margin (smoking rate) is the key analytical move. Generic “cigarettes are addictive” is too descriptive.

Question 4

[4 marks]

Price cap + subsidy: dual objectives.

Consumer protection. The S\$720 monthly fee cap at anchor operators directly limits the price parents pay, regardless of provider profitability. This protects low- and middle-income families from being priced out of childcare, particularly when anchor operators (PCF Sparkletots, MFS) account for the bulk of childcare market share. [2 marks.]

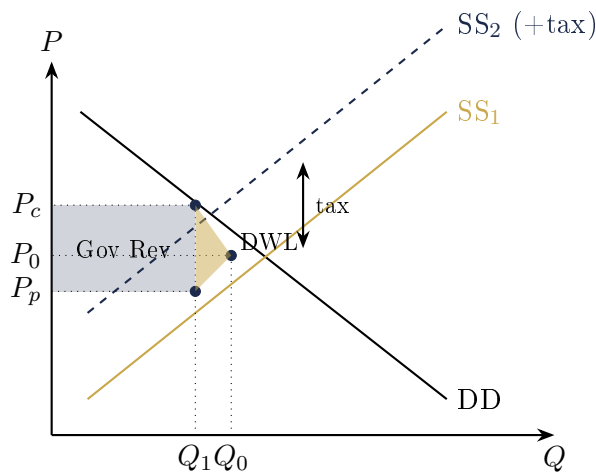
Merit-good correction. The S\$600 maximum subsidy further reduces the effective parental cost to S\$120/month for lower-income families. Combined with the operating subsidy to anchor operators (S\$1.26 billion in 2024), this raises consumption from below to closer to the socially optimal level Q^* , where $MSB = MSC$. Table 1.1 confirms the effect: childcare enrolment rose from 81% to 91% between 2018 and 2024 alongside the subsidy expansion. [2 marks.]

Examiner Note. Linking the dual mechanism back to the data (subsidy expansion + enrolment rise) demonstrates AO2 application. Pure theory without the data anchor caps at 3 marks.

Question 5

[6 marks]

Excise tax on tobacco: full welfare diagram.



The excise tax shifts the supply curve from SS_1 upward by the tax amount per unit to SS_2 . [1 mark for *SS shift*.]

At the original equilibrium price P_0 and quantity Q_0 , the tax creates a wedge between the price consumers pay (P_c) and the price producers receive (P_p). Consumers face a higher price $P_c > P_0$, and quantity demanded falls to Q_1 . Producers receive only P_p (the lower amount after the tax has been paid). [2 marks for P_c , Q_1 , P_p labelling.]

Welfare effects:

- *Consumer surplus* falls (consumers pay more for less quantity).
- *Producer surplus* falls (producers receive lower price on smaller quantity).
- *Government revenue* = (tax per unit) $\times Q_1$ = shaded rectangle.
- *Deadweight loss* (DWL) is the triangle between SS_1 and DD over $[Q_1, Q_0]$, reflecting the welfare loss from transactions no longer occurring.

[2 marks for full welfare decomposition.]

Because tobacco demand is price-inelastic (Extract 1.1), the quantity reduction (Q_0 to Q_1) is small relative to the price rise, so the consumer-surplus loss is concentrated in higher prices rather than lower quantity. This generates relatively large government revenue and relatively small DWL — the “efficient tax” property of taxing inelastic goods. [1 mark for *elasticity insight*.]

Examiner Note. The fully labelled diagram must show: SS shift, three prices (P_c , P_0 , P_p), two quantities (Q_0 , Q_1), government-revenue rectangle, DWL triangle. Missing any element costs marks.

Question 6

[6 marks]

Childcare subsidy: merit-good + labour-force.

Channel 1: Merit-good correction. Childcare generates positive externalities through skills accumulation and reduced future welfare costs. In a free market, the MPB undervalues these externalities, leading to under-consumption ($Q_m < Q^*$). The subsidy shifts the effective MPC down, raising consumption from Q_m toward Q^* . Extract 1.1 quantifies this: enrolment rose from 81% (2018) to 91% (2024) as the subsidy expanded. [2 marks.]

Channel 2: Labour-force participation. By reducing the parental cost of childcare to S\$120/month for lower-income families, the subsidy directly lowers the opportunity cost of female workforce participation. Mothers who would have stayed home to look after young children can now return to work at lower cost. Extract 1.1 reports that female labour-force participation rose from 60.4% (2018) to 64.5% (2024), consistent with this mechanism. [2 marks.]

Why these are complementary, not competing. The merit-good channel raises social welfare directly through children's developmental gains. The labour-force channel raises household income and aggregate AS (more workers, higher GDP). The two effects reinforce each other: working parents generate household income that supports the very children whose development is being subsidised. Critics note that the cost (S\$1.26 billion in 2024) is substantial, but the dual benefit makes the subsidy more efficient per dollar than single-purpose interventions. [2 marks: synergy + cost-effectiveness framing.]

Examiner Note. Top answers frame the policy as a *double dividend* (externality correction + labour-supply effect) rather than as a simple subsidy. Recognising the complementarity is the key analytical move.

End of Set 8 — Model Answers